Bohr's model or the Schrodinger's model. Similarly the subjective questions were also graded. The mean score of the students from the questionnaire in the section of structure of atom was found to be 5.16/6.00. The mean score for the section of dual nature of matter and radiation was found to be 3.70/6.00, $\sim 28\%$ lower than that seen in case of structure of atom. The mean scores of each section and the distribution of responses among them is summarized in Fig. 1.

After the commencements of the interviews, thematic analysis on the data was performed manually by the first author of the study from transcribed texts, following the six steps mentioned in [24]. Summary of the thematic coding can be found in Table I, table III and table IV.

Data from the responses of students was coded around the key concepts in the students' responses. The codes in Table II represent the sub-concepts that reflected themselves in their responses, and then the codes were centralized around the major themes. Table IV is a concurrent summary of both the questionnaire and the FG interview in terms of the frequency distribution of these themes among the students' responses.

Responses for the structure of atom spanned a number of models and concepts, starting from the classical analogy of the solar system to the quantum mechanical distribution of electron clouds around the nucleus. Students were also asked for their views around the absorption and emission of energy by an atom and the empty nature an atom. Following response is an example of how students reported their understanding and the coding of the same response in terms of, "revolving electrons", "centralized mass in atoms" and "probability distributions".

FG-student A: "Atom is mainly empty, three subatomic particle. Electrons revolve around nucleus in energy shells. Most of the mass of atom lies in nucleus."

FG-student B: "There are no such shells, but there is a region of high probability. There is no predefined shell that exist, it's just the high probability region that we call as shells."

The responses in the case of DoMR were a little less structured and were difficult to code to their central themes. The major ideas reported spanned from the experimental results of interference/photoelectric effect to the superposition of quantum mechanical states for the case of electron. Misconception, as used by Oslen (2002, [25]) is an umbrella term coined for the responses that were either incorrect statements or were major misconception around the concept of dual nature. These misconception were also reflected in the questionnaire as presented in Fig. 2(a) and Fig. 2(b). An example:

FG-researcher: "Does matter also have wave nature associated with them?"

FG-student: "Yes mam. For example us, when we do exercise our body gets heated and our body emits infrared waves or heat waves and that's how our body is related with waves."

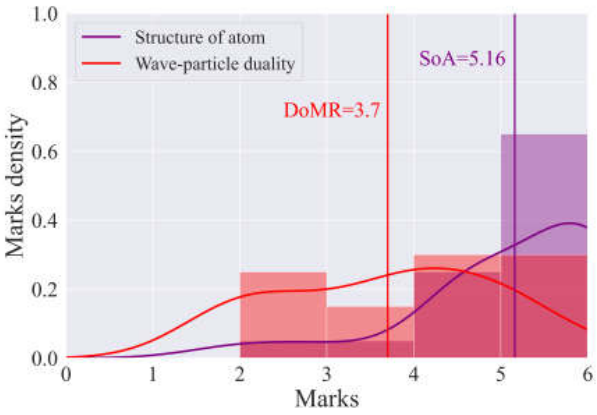


FIG. 1: The plot shows the kernel density estimate and the associated histograms of the marks density for the distributed questionnaire. The red (purple) plot and the histogram shows the marks distribution for the students for the section of DoMR (SoA). The mean score is shown with the vertical lines for each section.

Along with the conceptual questions and responses among the students, the focus group interviews were also structured to ask for their attitudinal responses and to reflect upon their experiences of learning these concepts. These responses were coded around the central factors that either fostered their interest in physics and positively shaped their learning experience or elements that were reported to be challenging or had an adverse effect on their learning. Another theme that came up often was their experience with learning and taking exams for the same. The attitudinal responses' thematic coding can be seen in Table II.

FG-researcher: "How did you feel learning about these concepts?"

FG-student A: "Mam, plum pudding model was easy but Bohr's model was a little difficult."

FG-student B: "Mam things started contradicting each other. For example a friend asked me if light is a solid or liquid or gas. And one friend asked me how does the transfer of the electrons happens?"

FG-student C: "Our known concepts were challenged all of a sudden. Electrons were particles, and light was a wave. Suddenly, both are both."

Sections	Themes	Codes
Structure of atom	Rutherford's model	∘ Revolving electrons ∘ Centrifugal forces
	Bohr's model	∘ Orbital Picture ∘ Quantization of momentum
	Schrodinger's model	∘ Electron Clouds ∘ Probability distributions ∘ Standing waves of electrons
	Atom is mostly empty	∘ Repulsive forces ∘ Concentrated mass at the centre
	Quantization of energy and angular momentum	∘ Quantization of energy ∘ Electron movement from one shell to another ∘ Energy absorption in discrete value
Wave-particle duality	Experimental results	∘ Interference and wave nature ∘ Photoelectric effect and particle nature ∘ Other Experimental evidences
	Misconceptions	∘ Incorrect statements ∘ Don't remember ∘ Empty responses
	Quantum mechanical picture	∘ Probability waves ∘ Superposition of states ∘ Particle nature and exertion of force
	Philosophical and personal opinions	∘ Neither wave nor particle ∘ Need for a better theory

TABLE I: Summary of the thematic coding for the conceptual understanding of the students: The thematic analysis was done following Braun, V. and Clarke, V six steps [24]. The accounts of the students were coded around the central themes of SoA and DoMR.

Sections	Themes	Codes
Attitudinal themes	Difficulties and negative influences	∘ Contradicting concepts ∘ Lack of experiments/hands on activities ∘ Not intuitive ∘ Mathematical difficulties
	Helpful stuffs positive influences	∘ Historical context ∘ Experimental exercises and activities ∘ Internet resources
	Exams	∘ Practicing questions ∘ Exam oriented teaching-learning ∘ Minimal contribution to conceptual understanding

TABLE II: Summary of the themes emerging out of the attitudinal responses: Students were asked a number of question to reflect upon their experience of learning SoA and DoMR. Their responses have been coded around the negative or positive influences each theme had on their learning.

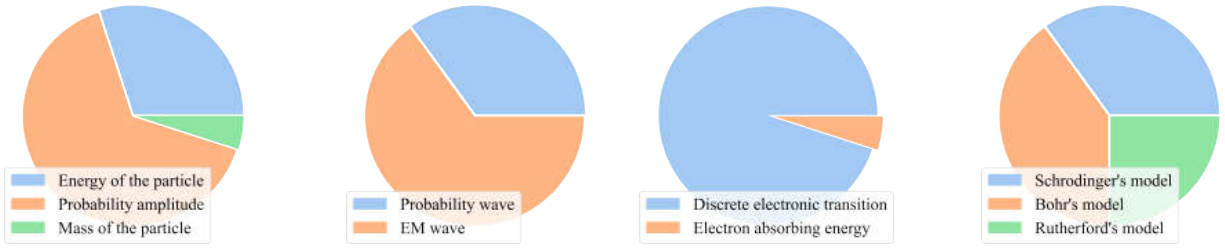
A. Analysis of students' response

This study was aimed towards investigating the students' understandings and difficulties around two concepts in modern physics, and reflect upon the differences in the learning because of the different ways they have been introduced and dealt throughout the school and undergraduate curriculum. The results from the questionnaire and the thematic analysis makes this difference evident.

The difference in the mean scores from the questionnaire (3.70 for DoMR and 5.16 for SoA, refer Table IV) reflects on the contrast in students' retention of the respective concept and their ability to reproduce the same. Fig. 1 summarizes the distribution of the scores of the student and shows that the associated kernel density estimate (KDE) in the case of SoA is skewed towards the higher end, while that of DoMR is rather flat, representing a consistently better performance in the case of SoA. Table III shows the distribution of individual student and their respective responses around the central themes. The table is a concurrent summary of the results from both the questionnaire and the focus group interview. The responses for the case of SoA is distributed

Students	Structure of atom					Wave-particle duality			
	Rutherford's model	Bohr's model	Schrodinger's model	Atom is mostly empty	Quantization of energy	Experimental results	Probability Wave	Misconceptions	Philosophical and personal opinions
Student 1	✓			✓	✓	✓		✓	
Student 2		✓		✓	✓	✘		✓	
Student 3			✓	✓	✓			✓	
Student 4		✓		✓				✓	
Student 5		✓		✓	✓	✓		✓	
Student 6	✓			✓	✓			✓	
Student 7		✓		✓	✓	✓		✓	
Student 8			✓	✓	✓	✓		✓	
Student 9			✓	✓	✓		✓		
Student 10	✓			✘				✓	
Student 11			✓	✓	✓			✓	✓
Student 12	✓			✓	✓	✘		✓	
Student 13			✓	✓	✓		✓		
Student 14	✓			✓	✓	✓		✓	
Student 15		✓		✓	✓		✓		
Student 16	✓			✓	✓			✓	
Student 17	✓			✓	✓	✘		✓	
Student 18		✓		✓	✓	✓			
Student 19			✓	✓	✓	✓	✓		✓
Student 20			✓	✓	✓	✓	✓		✓
Student 21		✓		✓	✓	✘	✓		
Student 22			✓	✓	✓	✘	✓		

TABLE III: Student wise analysis of the themes: The table is a combined analysis of the student's accounts from the questionnaire and the focus group interview. The students were consistent with their responses. Whenever a student has reported any account falling into any above mentioned themes or if their questionnaire responses adhered to that theme a ✓ has been marked, whenever a student reported the response and it was an incorrect response a ✘ has been marked. The empty spaces denote a case of no response from the student.



(a) What does the peak and trough of an electron wave denotes?

(b) What does the wave nature of a particle corresponds to?

(c) How does an atom emit or absorb radiation?

(d) Select the most appropriate description of the structure of atom.

FIG. 2: Pie-chart showing the distribution of answers across four selected questions from the questionnaire.

Section	Mean score
Structure of atom	5.16 (86%)
Wave-particle duality	3.70 (61%)

TABLE IV: The mean score (out of a total score of 6) of all the students from the questionnaire across the two sections

among different atomic models and almost every student reported a comprehensive understanding for the concentration of mass in the nucleus, quantization of the energy levels and the discrete nature of emission and absorption of energy by electrons. Following are some excerpts from the students which showcases the technical language compatibility among students around SoA:

FG-researcher: "Why doesn't the electron fall into the

nucleus?"

FG-student: "The orbitals they are in have constant energy, it doesn't gain or lose energy. Its not like a particle revolving, its a cloud of probability, revolution picture is not correct. It is only when we observe its here or there, otherwise it has an equal spread, it doesn't revolves in the modern picture."

It was seen throughout the FG interviews that even when asked about the structure of the atom, students would often mention quantum properties such as the probabilistic distribution of electrons, the quantization of electrons' angular momentum obtained by equating the wavelength to the circumference of the orbit, and the discrete nature in which energy is absorbed or emitted by electrons in the form of photons. Thus, inadvertently, students were seen to be discussing the concept of DoMR while talking about SoA.

On the other hand, the distribution of responses for the case of DoMR, in Table III, can be seen to be centralized around the experimental observation and misinterpretations. Only 7 students talk about the wave particle duality from a quantum mechanical standpoint, and only 3 of them have enough understanding to go further ahead to present their personal views into the topic apart from the textbook material.

FG-student: "Wave particle duality is taught by giving experiment and saying it cannot be explained by one so (it) must be the other. So it's just the way we are trying to explain (wave particle duality), both should be valid."

It was a common observation throughout the FG interviews that students were inclined to mention the experimental evidence for the wave-particle duality of electromagnetic radiation as soon as they were asked about the DoMR. Students didn't always have the retention as to which experiment highlighted the wave or particle nature, but their responses reflected their understanding of DoMR to be built around the experimental results. Students exhibited language compatibility around the experimental results, but when asked about further insights, their skewed conceptual understanding of the wave nature and the particle nature was evident. When asked "What do we mean when we say light has a wave nature?", almost 30% of the students reported that "light needs a medium to travel", along with a mention of the interference of light. Similarly, when asked to reflect on their understanding or visualization of the wave nature of particles, most of the students either had empty responses or major misconceptions about the topic. Fig. 2(b) shows how the majority of the students responded to electromagnetic waves when they were asked about what the wave nature of particles corresponds to. The results comply with the studies [25, 26] that suggest students' misconceptions are rooted in a classical physics worldview.

FG-student A: "I liked SoA. DoMR was not understood with clarity. We still don't understand how to see the waves or understand waves. We know how to calculate values but I don't really understand how does it happens."

FG-student B: "Light travels in a straight line, which also tells us about it's wave nature."

An important difference to note here is that while discussing SoA, students were seen to have a comprehensive understanding of the particle nature of waves and the wave nature of particles, as discussed earlier. Fig. 2(c) shows the percentage of students who mentioned the discrete nature of electromagnetic radiation, thus talking about the particle nature of waves. At the same time, when discussing these concepts in the realm of DoMR, students were seen struggling with the theory. Fig. 2(a) and Fig. 2(b) show the flawed understanding of DoMR among students. But more interestingly, when asked "What do the peak and trough of an electron wave denote?", about 65% of students answered that it is the probability amplitude, but when asked "What does the wave nature of a particle correspond to?", only 35% students answered probability wave, showing that there is a lack of consistency in their answers, reflecting an incomplete understanding of DoMR.

B. Attitudinal Responses

While collecting their attitudinal responses, students were asked to choose one topic that they felt was easy to understand. Of the 22 students, 13 chose SoA, 4 chose DoMR, 1 student said none as they found both on equal footing, and 4 students didn't respond. The thematic analysis of the attitudinal responses reflects the factors that might have had a positive or negative influence on the learner. One major factor that came up too often was how the historical and contextual information helps the students not only relate to the concept but also construct the concepts in their minds around its needs, discoveries, experiments, and conclusions. [27, 28]

FG-student: "There was a curiosity to understand SoA. Also the history that was told about the structure of atom helped a lot in understanding them. It started very basic."

The majority of the students mentioned that they needed explicit help, either in the form of other books, internet searches, or video explanations, to understand DoMR when having difficulties. In the entire sample space, 4 students reported still being curious about these concepts and either pursuing their career forward in this avenue or learning about these concepts independently.

FG-researcher: "Do you think learning about these con-

as textbooks for the undergraduate curriculum by UGC. The coursework begins with an introduction of the failures of classical theory by presenting some key experimental results showcasing the particle nature of electromagnetic radiation: Planck's law on black-body radiation and Einstein's theory of the photoelectric effect. This is followed by a discussion on the Compton scattering of electrons, introducing the wave nature of particles. With the experimental results presented, the uncertainty principle and the DoMR is introduced. A brief discussion is presented regarding the consequences of the DoMR. These postulates are then used to motivate the development of the Schrodinger's equation. Students are introduced to different techniques to solve the Schrodinger's equation for simple quantum mechanical systems. After the introduction of the basic postulates of quantum mechanics, the different models for the SoA are introduced and discussed in detail. The curriculum concludes with the calculation of the energy eigenstates and the electron wavefunction of simple atoms such as that of the Hydrogen atom and discussing the implications of the existence of discrete energy eingenstates.

The syllabus summary is presented in Table V. The table includes the topics covered with respect to their time of introduction in the curriculum, subject, and context. Till 10th grade, students do not have the segregation of science as physics, chemistry, or biology. It happens in the 11th standard, only for those who chose the science stream.

SoA is dealt with in a structured and gradual fashion, enriched with the historical context of its development. From its first introduction in the 9th standard to the quantum mechanical discussion in the 12th standard, it follows a structured and repetitive pattern. On the other hand, the introduction of DoMR is rather brief and abrupt. Also, the context of the very first introduction of the wave nature of particles follows a very classical approach to sound waves. After that, the wave theory gets its place in a scattered and nonstructured way. The first quantum mechanical description of DoMR in the 12th standard seems abrupt and all at once. There's a profound difference in the curriculum about both concepts in terms of their first introduction, context, historical progression, and experimental proofs.

VI. DISCUSSION

The results from the FG interviews and questionnaires can be very well understood if we put them in perspective with the curriculum around these concepts. The wave-particle duality was introduced in the school syllabus in the 12th standard. The learning begins with a dedicated chapter on interference and diffraction, which discusses the wave nature of light in detail. This is followed by a discussion about the photoelectric effect in detail, discussing the particle nature of light. Experimental results are mentioned in detail, but there is very little or no elaboration on the facts about what it really means to have a wave or particle nature and how they contradict each other. In the earlier standards, the concept of wave is introduced in the context of sound waves and how they need a medium to travel. The drastic transition to electromagnetic waves and their wave/particle duality are not dealt with carefully, leaving a void for the student to make their own sense of these concepts, resulting in misconceptions or leaving unanswered curiosities, resulting in either empty or no responses.

On the other hand, the SoA has remained a recurring concept in the curriculum since the 9th grade. The introduction in the 9th standard gives them comprehensive knowledge about the atomic model, building the concepts of centralized mass and electrons being in certain "shells" or "orbits". The subject is re-introduced in the 11th standard, with a more detailed description of its quantum mechanical character. But despite the recurring nature of the concept, the fact that 14 out of 22 students responded with either the classical Rutherford's model or Bohr's model supports the constructivist hypothesis in the early years of a student's career.

FG-student: "We keep discarding the different models of SoA but the Bohr's model wasn't discarded at that moment which made us believe Bohr's atomic model is the ultimate truth and it sticks with us for longer times. We all have studied about atomic clouds and probability distribution but still many of us imagine the Bohr's model for SoA because it retained very much"

Students didn't report the SoA being very difficult, as the concepts were not contradictory but rather constructing around the structure of atoms. On the other hand, putting the wave particle duality later in the curriculum takes away the opportunity for construction and lays down the foundation for cognitive conflicts and contradictions around the concerned concept.

Based on the above results, we propose the following improvements and ways to deal with the subject matter in a more comprehensible fashion:

- The early introduction of the concepts of modern physics will help in the construction of knowledge in an accurate and more comprehensible way. Throughout the study, the socio-cultural constructivism theory of learning has presented itself in many shapes and forms. The early years of a child's education contribute to the construction of their worldview, which will manifest throughout their lives. Hence, this should be the time for the introduction of the most appropriate understanding of the physical world. They should be taught to appreciate and take benefit of the historical developments that led to the destination, but they need not take the same journey.

Curriculum	Standard	Structure of atom	Wave-particle duality
School curriculum	Standard 9	∘ Thomson's model ∘ Rutherford's model ∘ Bohr's model ∘ Electrons in shells (K,L,M,N..) ∘ Valency	—
	Standard 10	—	—
	Standard 11	(Chemistry) ∘ Schrodinger's model ∘ Pauli's exclusion principle ∘ Quantization of momentum ∘ Different energy levels and electron's movement in that	(Chemistry) ∘ Dual nature of matter and radiation discussed very briefly in the chapter for SoA
	Standard 12	(Physics) ∘ Electron Clouds ∘ Probability distributions ∘ Standing waves of electrons ∘ Quantization of energy	(Physics) ∘ Wave nature of Light ∘ Interference and diffraction ∘ Photoelectric effect /particle nature of light ∘ Davission and Germer experiment /wave nature of electrons ∘ Interference of light and matter and the understanding of wave nature associated ∘ Photoelectric effect and particle nature of light and electrons ∘ Experimental evidences reflecting the need of dual nature
Undergraduate curriculum	1st year	∘ Review of atomic models ∘ Bohr's model and postulates ∘ Sommerfield's model ∘ Solution of one electron systems ∘ Energy eingenstates and wavefunction interpretation of H-atom	∘ Discrepancy with black body spectrum and photoelectric effect ∘ Planck's theory of black body ∘ Einstein's theory of photoelectric effect ∘ Compton effect ∘ Matter waves ∘ Dual nature of matter and radiation

TABLE V: The list of subject matter and topics, and when they have been introduced in the curriculum. For school syllabus we have referred to the syllabus prescribed by the CBSE with NCERT textbooks. The undergraduate syllabus has been taken from the UGC prescribed syllabus for the undergraduate physics course.

- Concepts should be introduced with their most accurate description, simplified, and tailored to their age-appropriate cognitive development. EF project has repeatedly shown this in several studies over the past 10 years [4]. The curriculum and specific lesson plans could be curated to make the content cognitively competent for the learner. The focus should be not to overburden them but, also not to introduce them to older or less accurate scientific content.

- Hands-on activities and appropriate models should be used in the visualization of these concepts. Online resources are already helping students. Conscious and careful development of such online resources will significantly improve teaching and learning. As students have reported taking help from resources other than the textbook and course material, in this new age of technology we should take advantage of the vast range of resources that can be developed to aid the visualization of the rather abstract concepts [32–36]. ReleQuant has developed an array of such open online resources for several concepts in modern physics. Likewise, these multimedia resources must be combined with research-based teaching and learning strategies [37].

VII. CONCLUSION

We live in an era where the exponential growth of science has been so normalized that nothing seems impossible anymore. Children don't see the awe in stepping on the moon, and knowing the location of your loved ones with the vast network of satellites has become a necessity. And yet, modern-day understanding in the commons about the universe around them lacks the depth and accuracy that are needed for the next generation to take science forward and appreciate the nuances and beauty of the world that we live in.

The study aimed to answer the questions that revolved around the understandings and conceptions of two modern physics concepts among science students by assessing the motivations and challenges from the perspective of the school and undergraduate physics curricula. A con-

trast between two important modern-physics concepts, the structure of the atom and the dual nature of matter and radiation, was proposed because of the early and recurring introduction of the one while brief and abrupt discussion of the other.

With a combination of a questionnaire and semi-structured focus group interviews, the study glides into students' understanding and reflects the disparity between the two: SoA and DoMR. The relation between their alternative conceptions and the treatment of the concepts in the curriculum, emerges from the thematic analysis conducted on students' accounts from the interview.

The curriculum of physics in India, for senior secondary students, does talk about quantum mechanical concepts such as the Schrodinger model of the structure of an atom, the dual nature of matter and radiation, and Heisenberg's uncertainty principle. However, a contrast study of the two topics reflects the disparity between the treatment in the secondary school curriculum. On one hand, the SoA was introduced in the 9th standard with a qualitative description of the historical developments of different models, ending its note by presenting Bohr's model. The chapter takes the reader on a historical journey by discussing major scientific milestones, such as the discovery of subatomic particles, in the context of the development of SoA. The later section presents a vague idea of electron shells and the filling of electron shells with the rule of $2n2$. Students' accounts from the interviews and their responses to the questionnaire show that the curriculum reflects itself in their understanding. Although students were introduced to the modern-day Schrodinger model in their senior secondary curriculum, most students tend to retain Bohr's model or the classical Rutherford's model, as it was the last model taught to them in their secondary school years.

DoMR has not been treated with appropriate care and detail. Students were seen to report classical interpretations of the wave nature of radiation. The students did talk about the experimental evidence of the photoelectric effect and interference to mention the origin of the problem, but when asked further open-ended questions about the wave nature of particles, misconceptions in their understanding were pretty evident.

The reason for this could be attributed to the fact that the curriculum might consider the topic cognitively challenging for the students, hence not giving it the needed attention. But at the same time, SoA, also a quantum mechanical and non-trivial concept, gets its needed attention, and students can be seen showing their compatibility around the concept.

Students are able to reproduce textbook accounts of DoMR, but when asked further open-ended questions, they can be seen to struggle and report major misconceptions. The students had difficulty understanding the wave and particle nature of matter and radiation, which reflected the lack of connections around the concept. On the other hand, most students can be seen as more comfortable and responsive around the SoA, reflecting the normalization of the concept in the socio-cultural setting.

The results support the constructivist hypothesis of Vygotsky that the students construct their knowledge through their interactions with the environment. The concepts, taught with needed care and attention in the early years of their schooling, get constructed in their minds as such, and the children make sense of them naturally. Subject matter taught in the early years of schooling makes their world of common sense difficult to challenge later in their lives. Most of the students are still reporting Bohr's model, although all of them have been introduced to the quantum mechanical model, which is evident of this very fact.

This study proves that if students are capable of understanding the SoA, they are also capable of understanding modern-day physics concepts provided the needed attention to the subject matter in the curriculum. We anticipate more studies to follow in the Indian context to develop interactive online and offline curricula to teach modern physics concepts and to see the applicability of the same in classrooms. We believe that students deserve the best understanding of the contemporary world, and we need to work towards this.

ACKNOWLEDGMENTS

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APPENDIX A: FOCUS GROUP INTERVIEW GUIDE

- **Conceptual Understanding:** What do you understand by the wave particle duality? What does the wave nature of light correspond to? What do you think experimentally prove the wave nature of light and how? How do you visualise an electron having a wave nature?

  Can you explain the structure of an atom to me? Why doesn't the electron fall in the nucleus? How does an atom absorb energy? What happens when it absorb energy? How is an atom of a calcium different from that of an iron? Does every material around us have atoms as their building blocks? Do you think the living organisms are also made up of atoms?